

Supplemental Material for: Phonons in potassium doped graphene: the effects of electron-phonon interactions, dimensionality and ad-atom ordering

(Dated: August 24, 2011)

I. CHARACTERISATION OF UNDOPED 1-4 LAYER GRAPHENE

Graphene samples were prepared by micromechanical exfoliation of natural graphite onto an oxidized Si substrate. In order to investigate the effect of doping on 1-4 layer graphene, a substrate was fabricated which contained all four samples to enable comparison of Raman

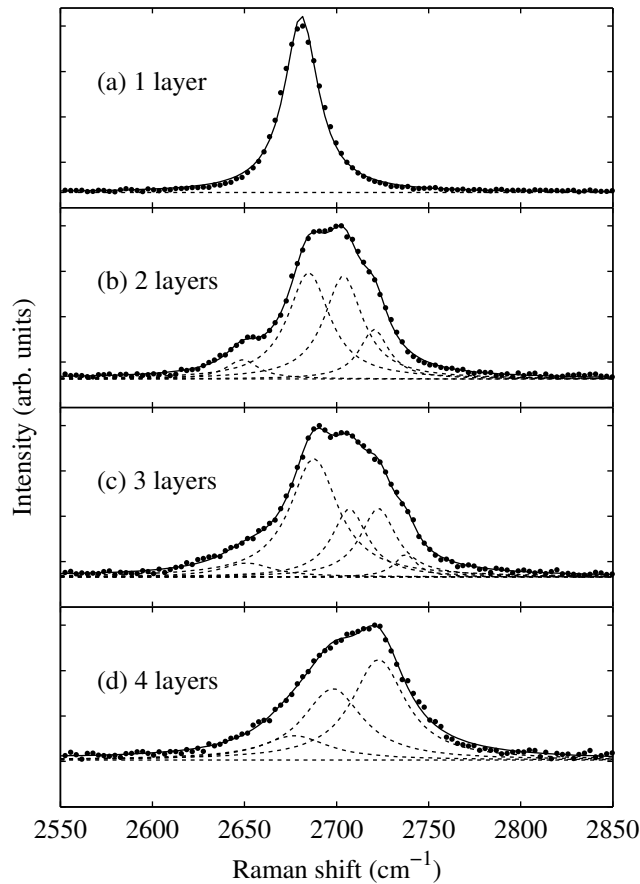


FIG. 1. The 2D band of the undoped 1-4 layer graphene samples which were studied as a function of doping: (a) 1 layer, (b) 2 layers, (c) 3 layers, (d) 4 layers. Solid lines are fits to the data, composed of the peaks shown by dotted lines.

spectra at identical exposures to potassium vapor. Graphene samples were identified by optical contrast and Raman spectroscopy [1]. To analyze the relative intensity and lineshape of the Raman peaks, we fitted the data using a Lorentzian function for the 520 cm^{-1} silicon peak and 1583 cm^{-1} graphite G-peak. The graphene 2D-peaks were fitted with the sum of multiple Lorentzian components.

It has been shown that the shape and intensity of the 2D peak of few layer graphene is strongly dependent on the number of layers in the graphene stack [1, 2]. Fig. 1 shows the fitted 2D peak for 1-4 layer graphene used in this work, which is in close agreement to the known lineshapes for 1-4 layer graphene [2].

Further confirmation of the number of layers in the sample is given in Fig. 2, where we examine the integrated intensities of the G-peak and the Si 520 cm^{-1} peak. The relative intensity of the G peak compared to the Si peak, as plotted in Fig. 2(e), increases with the number of graphene layers as one might expect. Furthermore, the intensity ratio closely follows the trends identified in Ref. [3], which presents a detailed study of this method for counting the number of graphene layers. This trend, combined with the changes in the

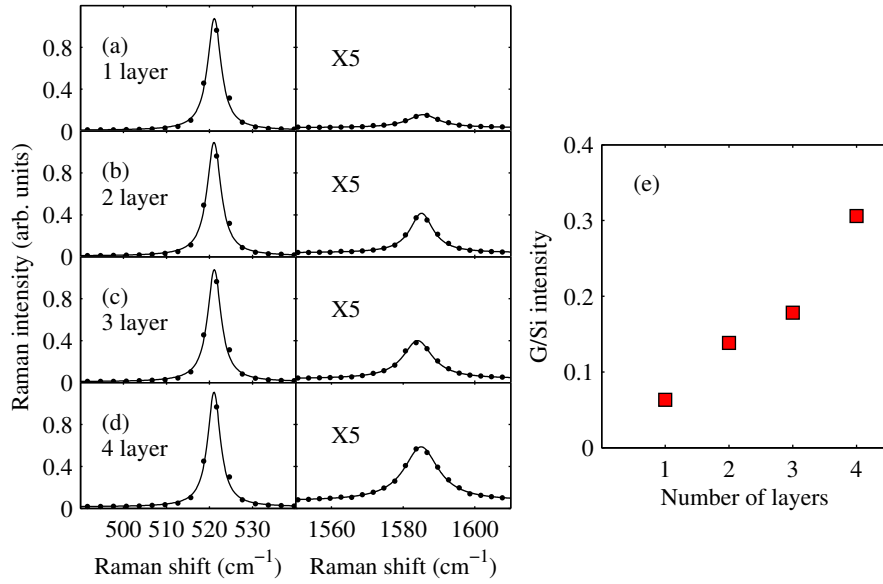


FIG. 2. The changing ratio of the integrated intensity of the G-peak to the 520 cm^{-1} Si peak. (a)-(d) show fitted the Si and G-peaks for graphene thickness 1-4 layers, (e) $I(G)/I(Si)$ as a function of designated number of layers. The intensity of the G peak has been magnified by a factor of 5 so that it can easily be seen on the same intensity scale as the Si peak.

lineshape of the 2D mode, provided a convincing verification of the number of layers in our graphene samples.

II. IDENTIFYING MANY LAYER GRAPHENE

To investigate the crossover in behavior between K-graphene and K-graphite, we isolated a relatively thick graphene sample containing tens of graphene layers. The number of layers can be estimated by analyzing the color of the graphene sample in an optical micrograph. When light passes through the air-graphite-SiO₂-Si stack, interference between the different paths of the light changes the reflectance spectra. Studies of how the (red-green-blue) RGB color scale changes with flake thickness have been previously reported [4]. Figure 3 applies this analysis to our graphene flake, where we determine RGB values of (0.43,0.36,0.38). This corresponds to a thickness of about 100 graphene layers.

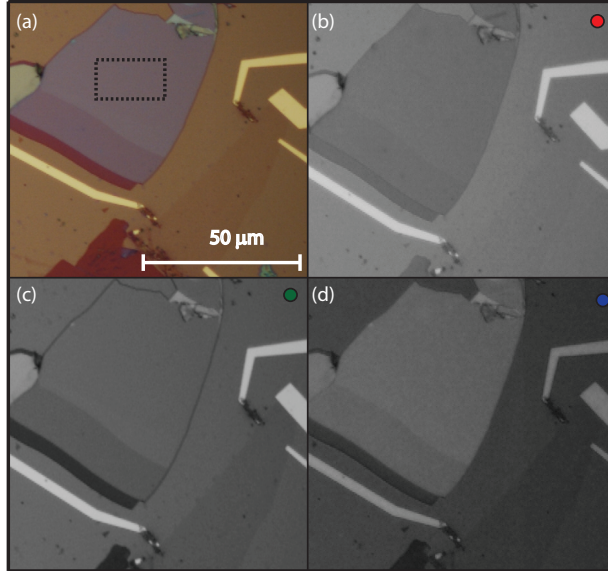


FIG. 3. (a) Optical micrograph of the thick, bluish many layer graphene flake (identified by the dotted rectangle), (b)-(d) show the same image split into (b) red (500-650 nm), (c) green (450-600 nm), and (d) blue (400-500 nm) component images.

III. FITTING TO THE BREIT-WIGNER-FANO LINESHAPE

The G-peak of the K-doped graphene spectra is fitted with the asymmetric Breit-Wigner-Fano lineshape. This lineshape arises due to the interference between the phonon (a discrete energy state) and an electronic continuum [5] and is commonly found in the Raman spectra of doped graphitic systems [6]. It is modelled as a signal of intensity:

$$I(\omega) = I_0 \frac{(1 + \frac{\omega - \omega_{ph}}{q\Gamma/2})^2}{1 + (\frac{\omega - \omega_{ph}}{\Gamma/2})^2}. \quad (1)$$

Here ω_{ph} and Γ are fitting parameters to the central frequency and full width at half maximum (FWHM) of the bare phonon. The $1/q$ factor is the Fano parameter which describes the nature of the interference between the phonon and the electronic continuum, and the asymmetry of the resulting lineshape. If $|1/q| \gg 1$ transitions to the electronic continuum dominate over the phonon and there is an anti-resonant dip in the spectrum; if $|1/q| \ll 1$ the phonon dominates, and there is a resonant peak in the spectrum. In the limit $1/q \rightarrow 0$ a Lorentzian lineshape is recovered. When fitting this lineshape to the data, we typically chose a data window between 1200 and 1800 cm^{-1} and the feature at $\sim 1260\text{cm}^{-1}$ was fitted with a Lorentzian. This reasonably simple model accurately fits the data over our entire doping range, over which the spectrum changes considerably, as shown in the example fits in Fig. 4. The fitting was rigorously checked by confirming that similar results were obtained when changing the windows around the peak. In the Raman experiments, a significant amount of continuum scattering was observed. While some of this may result from electronic Raman scattering from the sample, much of this results from the substrate and the sample container. Consequently, the intensity of this background depends on the path of the light and the focusing in the Raman experiment. The background was approximated by a straight line in the window of the fit. We investigated the validity of this approach by confirming that the fitting results are independent of the windows of the data. We also obtain consistent results on different samples, with the light going through different regions of the sample container.

Although we always measure a negative $1/q$ of -0.2 — -0.3, indicating a phonon-dominated response with some coupling to the electronic continuum, we found no clear trend of this parameter with increasing doping. This is perhaps not surprising as the Fano parameter depends on the interference between the laser, the phonon and the electronic continuum, and is therefore expected to change with high doping in a non-trivial way.

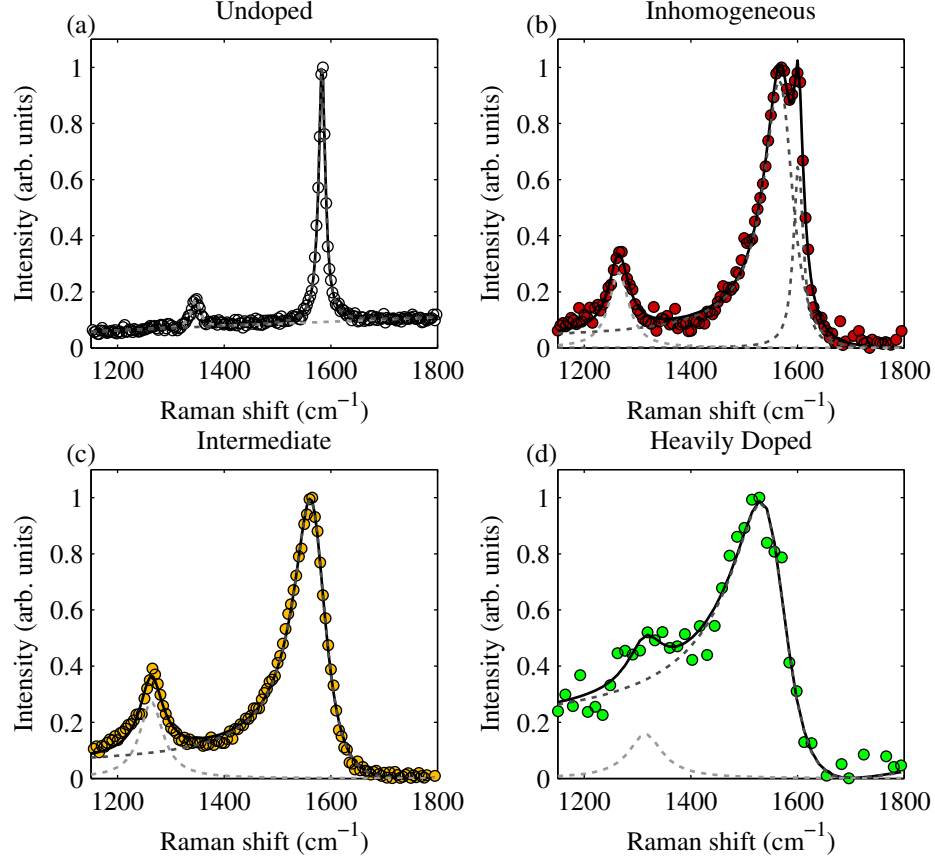


FIG. 4. Example fits of the G-peak in the different doping regimes: (a) undoped, (b) inhomogeneously, (c) intermediately, and (d) heavily doped graphene. Solid lines are fits to the data, composed on the different peaks shown by dotted lines.

-
- [1] A. Ferrari, J. Meyer, V. Scardaci, C. Casiraghi, M. Lazzeri, F. Mauri, S. Piscanec, D. Jiang, K. Novoselov, S. Roth, and A. Geim, *Phys. Rev. Lett.* **97**, 187401 (2006).
 - [2] L. Malard, M. Pimenta, G. Dresselhaus, and M. Dresselhaus, *Physics Reports* **473**, 51 (2009).
 - [3] Y. K. Koh, M.-H. Bae, D. G. Cahill, and E. Pop, *ACS Nano*. **5**, 269 (2011).
 - [4] S. Roddaro, P. Pingue, V. Piazza, V. Pellegrini, and F. Beltram, *Nano Letters* **7**, 2707 (2007).
 - [5] U. Fano, *Phys. Rev.* **124**, 1866 (1961).
 - [6] M. S. Dresselhaus and G. Dresselhaus, *Advances in Physics* **51(1)**, 1 (2002).